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MILITARY SPECIFICATION

BREATHING APPARATUS, OXYGEN GENERATING, TYPE A-4 (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-generating, type A-4 oxygen breathing apparatus for use by firefighting, damage control, and other personnel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-393	- Plastic Molding Material, Polycarbonate, Injection and Extrusion.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-17671	- Canister, Oxygen Breathing Apparatus, for Type A-4.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-151	- Metals: Test Methods.
FED-STD-191	- Textile Test Methods.
FED-STD-595	- Colors.
FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-2073-1	- DoD Material Procedures for Developing and Application of Packaging Requirements.

2.1.2 Other Government drawings and publication. The following other Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5184270	- Oxygen Breathing Apparatus Navy Type A-4, Assemblies and Details.
803-5184258	- Oxygen Breathing Apparatus, Canister.
804-5773272	- Lockers, Oxygen Apparatus Shelf Unit.

PUBLICATION

NATIONAL BUREAU OF STANDARDS (NBS)

Circular Special 374 - Methods of Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Standard Test Methods for Rubber Properties in Tension. (DoD adopted)

D 1003 - Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Composition. The material shall be as specified herein and in Drawing 803-5184270. Metal parts shall be corrosion-resistant or coated to protect them from corrosion. Elastomeric parts shall be resistant to deterioration caused by ocean environment and shall be nonirritating to the user of the apparatus.

3.2.2 Rubber compound. The rubber compound shall be 70 percent natural rubber and 30 percent nitrile. The nitrile component shall be a copolymer of butadiene and acrylonitrile. The rubber shall contain no mineral sulphate or lead. It shall contain an antioxidant compound equivalent to those known as VGB or Agerite White. The rubber compound shall show no irritation to the skin (see 4.6.1.2). The finished rubber shall be black in color in accordance with FED-STD-595 and shall have the characteristics shown in table I (see 4.6.1.1).

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TABLE I. Physical characteristics of rubber compound.

Test	Before aging	After aging (percent change)
Tensile strength	1.758 X 10^6 kg/m^2 (3.600 X 10^5 lb/ft^2)	+20 -25
Ultimate elongation (percent, min)	550	+20 -15
Permanent set[1]	20	+20 ---

[1] After elongation to 75 percent of the ultimate elongation for 10 minutes and resting for 10 minutes.

3.2.3 Neoprene-coated fabrics. The rubber compound used in the fabric shall be a polymer of chloroprene (2-chloro-1, 3-butadiene). The coated fabric shall be made of cotton and shall be in accordance with Drawing 803-5184270. It shall be black in color in accordance with FED-STD-595 and the neoprene shall have the physical characteristics shown in table II (see 4.6.1.1).

TABLE II. Physical characteristics of neoprene rubber.

Characteristic	Before aging	After aging
Minimum tensile strength	1.14 X 10^6 kg/m^2 (2.340 X 10^5 lb/ft^2)	8.437 X 10^5 kg/m^2 (1.728 X 10^5 lb/ft^2)
Ultimate elongation (minimum) (percent)	750	635

3.2.4 Self-extinguishing material. The coated fiberglass cloth that shall be in accordance with Drawing 803-5184270 shall flame out within 2 seconds (see 4.6.8). The exposed side of the cloth shall be black in color in accordance with FED-STD-595.

3.2.5 Threads. Threads shall be in accordance with FED-STD-H28 and Drawing 803-5184270.

3.2.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

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3.3 Assembly. The breathing apparatus shall be a closed circuit and self-contained in accordance with Drawing 803-5184270. The apparatus shall ensure the maximum safety and comfort of the wearer in accordance with MIL-STD-1472.

3.4 Construction and maintenance. Any part of the apparatus that is in direct contact with the wearer's body shall be of nonirritating composition. The apparatus shall contain the following principal parts:

(a) Mask assembly
  (1) facepiece
  (2) facepiece form
  (3) lens
  (4) lens cover/protector
  (5) lens retaining ring
  (6) nosecup assembly
  (7) speaking diaphragm assembly
  (8) speaking diaphragm wrench
  (9) combination valve
  (10) head harness
  (11) buckles
  (12) breathing tubes
(b) Body assembly
  (1) plunger assembly
  (2) canister guard and breastplate
  (3) retaining assembly
  (4) breathing bags
  (5) relief valve assembly
  (6) timer assembly
(c) Harnesses
  (1) waist belt
  (2) harness assembly

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.7.

3.5.1 Breathing resistance. Resistance to gas flow of an apparatus complete with a dummy canister shall not exceed 75 millimeters (mm) (2.95 inches) of water column height (see 4.6.3).

3.5.2 Leakage. There shall be no leakage in excess of the amounts specified in 3.6.1 for the facepiece assembly, 3.6.2 for the body, and 3.6.2.2.1 for the relief valve assembly (see 4.6.9 through 4.6.9.3).

3.5.3 Corrosion resistance. Immediately after exposure to the salt spray atmosphere specified in 4.6.4, the apparatus shall have no visual sign of corrosion on the operating mechanism, canister guide, or facepiece.

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3.5.4 High impact shock. The apparatus shall meet the leakage requirements of 3.6.1 and 3.6.2 (see 4.6.5).

3.5.5 Impact. The apparatus body shall be dropped as specified in 4.6.6 and shall be fully operable with a facepiece attached.

3.5.6 Wear. Repeated operation of the bail as specified in 4.6.7 shall not degrade the operation of the apparatus.

3.5.7 Apparatus output. The apparatus, when used in conjunction with a canister which is in accordance with MIL-C-17671, shall supply breathable gas, as defined in MIL-C-17671, for a 50-minute minimum. The temperature of the breathable gas shall not exceed 46 degrees Celsius (°C) (115 degrees Fahrenheit (°F)) (see 4.6.10). The average carbon dioxide volumetric content of the gas received from the apparatus shall not exceed 1.5 percent nor shall the carbon dioxide volumetric content of the gas exceed 5.0 percent at any time (see 4.6.10).

3.6 Components. Leakage requirements shall be as specified in 3.5.2 (see 4.6.9 through 4.6.9.4).

3.6.1 Facepiece assembly. The facepiece assembly shall be in accordance with Drawing 803-5184270. It shall include the facepiece, hoses, and valves necessary to bring breathable gas from the body of the apparatus to the respiratory passages of the wearer and to conduct exhaust gas back to the body of the apparatus. The allowable leakage shall be 2 percent by volume, during the 30 seconds after the test has started (see 4.6.9). The components of the facepiece assembly shall be in accordance with Drawing 803-5184270 and the following as specified (see 4.6.9.2).

3.6.1.1 Facepiece. The facepiece shall be made of molded rubber as specified in 3.2.2. The facepiece shall have one lens.

3.6.1.1.1 Facepiece form. A plastic facepiece form in accordance with Drawing 803-5184270 shall be installed in each facepiece to ensure protection of the facial seal during storage.

3.6.1.2 Lens. Each facepiece shall be fitted with a lens molded in accordance with Drawing 803-5184270 from a colorless polycarbonate resin in accordance with L-P-393. The lens shall not crack, fracture, chip, or dislodge from the retaining rings (see 4.6.2.3).

3.6.1.2.1 Optical properties. The finished lens shall be free of striae scratches, discoloration, or other defects interfering with clearness of vision or causing eyestrain under prolonged use. Standards in this respect are those used for ophthalmic lenses and not those used for high precision optical devices. The lens shall transmit not less than 85 percent of light incident on the surface. The lens shall not be positive in refractive power and shall not have a negative refractive power greater than 1/4 diopter (see 4.6.2.4).

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3.6.1.2.2 Lens construction. The outer cylindrical surface of the lens shall be molded to the radii as specified on Drawing 803-5184270. The inner surface shall be molded to a cylindrical form calculated to accomplish an error refraction within the limits stated in table III. These limits shall apply to all points on the lenses inside a line, parallel to the axis of curvature, and subtending at an angle of 37 degrees with the optical axis (see 4.6.2.2).

TABLE III. Lens construction details.

Cylindrical powers[1]	Convexity zero	Concavity 1/4 diopter (max)
Prism power, vertical	Superconducting 1/7 diopter	Subducting 1/8 diopter
Prism power, horizontal	Abducting zero	Abducting 1 to 1-1/2 diopters

[1] These limits are applicable to all points on the lens inside a line parallel to the periphery of the lens, which is a distance of 12.7 mm (0.5 inch) from the edge.

3.6.1.3 Lens cover and protector. A self-adhesive lens protector shall be provided to protect the lenses during shipping. The protector shall not degrade lens performance below the requirements of this specification. The protector shall be easily removable.

3.6.1.4 Lens retaining rings. The plastic lens shall be retained in place in the molded opening in the facepiece by means of lens retaining rings. The retaining rings shall be of a molded material and made so that when in place they shall compress the rubber under the solid part of the retaining ring forming a gas-tight seal.

3.6.1.5 Nosecup assembly. A nosecup assembly shall channel the moist, exhaled gas to the exit port in the combination valve. The nosecup assembly shall be installed so that it will not restrict breathing or visual lines of sight.

3.6.1.6 Speaking diaphragm assembly. The facepiece shall be equipped with a speaking diaphragm assembly.

3.6.1.6.1 Diaphragm wrench. A wrench for removing the cover over the speaking diaphragm shall be provided in the shipping container.

3.6.1.7 Combination valve. The facepiece shall be equipped with a combination valve box that shall direct the flow of exhalation and inhalation gas. The valve shall be of a sturdy leakproof construction, shall direct breathable gas from the body of the apparatus toward the wearer, and exhaust gas to the body of the apparatus.

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3.6.1.8 Head harness. The finished head harness shall be made of molded rubber in accordance with 3.2.2 and shall be molded not more than 12 months before delivery. The head harness shall be adjustable and shall maintain tension for 1 hour. The rubber of the head harness shall be black in color in accordance with FED-STD-595 and shall have physical characteristics as specified (see 3.8 and 4.6.1.1).

3.6.1.8.1 Buckles. The facepiece shall be equipped with buckles in accordance with Drawing 803-5184270. The buckles shall be compatible with the head harness straps, shall be constructed of stainless steel, and shall be the quick-adjustable, roller type.

3.6.1.9 Breathing tubes. The facepiece shall be equipped with two breathing tubes, one for inhalation and one for exhalation. The breathing tubes shall be constructed from rubber as specified in 3.2.2 and shall be gas-tight (see 4.6.11).

3.6.1.9.1 Breathing tube fittings. The ends of the breathing tubes shall terminate in quick-connect fittings that match those on the body of the device. The inhalation and exhalation hoses shall have different size fittings to prevent the wearer from attaching the hoses to the wrong ports during operation of the device.

3.6.2 Body. The body of the apparatus shall be in accordance with Drawing 803-5184270. The body shall include a canister holder, breathing bags, and a timer. It shall purify and oxygenate the wearer's breathing gas. The allowable leakage of the body (see 4.6.9) shall be 3.05 centimeters (cm) (1.2 inches) of water column height within the first minute of measurement. The components of the body shall be in accordance with Drawing 803-5184270.

3.6.2.1 Canister holder. The canister holder shall consist of the plunger assembly, canister guard, breastplate, and retaining system. In accordance with Drawing 803-5184258, these parts shall be assembled to permit the canister to be inserted and held in a standby position without puncturing the canister seal. Operation of the retaining assembly shall ram the canister home on the plunger assembly, piercing the canister seal and shall gas-tight seal the canister to the apparatus at the plunger assembly.

3.6.2.1.1 Plunger assembly. The canister holder shall be equipped with a plunger assembly that shall pierce the canister seal in accordance with MIL-C-17671.

3.6.2.1.2 Canister guard and breastplate. The canister holder shall be equipped with a canister guard and breastplate that shall be covered with an insulating, self-extinguishing (see 3.2.4) material on all externally exposed surfaces.

3.6.2.1.3 Retaining system. While in the standby position, the retaining system shall allow easy insertion of the oxygen breathing apparatus canister in accordance with MIL-C-17671. The retaining system shall have a handle that shall latch securely in the standby position to retain the canister without

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puncturing the (copper foil) seal (see 4.6.6). Movement of the handle to the latches (operational) position shall lift the canister to a seated position with the plunger housing. The canister spring (in the retaining system) shall exert sufficient force to prohibit leakage between the canister and the plunger housing (see 4.6.9).

3.6.2.2 Breathing bags. The breathing bags shall be constructed of neoprene coated fabric and shall have the physical characteristics as specified in 3.2.3. The bags shall be impermeable to air at pressures up to 25.4 cm (10 inches) of water column height (see 4.6.9.1). The breathing bags as installed in the apparatus body shall have a capacity of 5 liters (0.18 cubic feet (ft^3)) minimum (see 4.6.14).

3.6.2.2.1 Relief valve assembly. The left breathing bag shall be fitted with a relief valve. The valve shall operate by means of an internal mechanical linkage when the bag expands. The valve shall incorporate a check valve to prevent entry of ambient air into the apparatus. The relief valve shall be spring loaded with sufficient tension so that when in the closed position, leakage will not exceed 0.10 milliliter per minute (mL/min) (0.0061 cubic inch per minute (in^3/min)) (see 4.6.9.3).

3.6.2.3 Timer assembly. A timer assembly shall be installed on the body of the apparatus. The timer shall sound a warning signal in the form of a continuous ring of 10 seconds duration (minimum) (see 4.6.12.2). This signal shall operate after the timer has been fully wound and shall be set for any time period up to 1 hour. The timer shall be spring activated, and the spring shall be made of materials resistant to salt water corrosion. One control knob shall wind, set, and activate the timer. The timer shall not deviate more than plus or minus 5 minutes in 1 hour. The sound intensity of the ring shall be 75 decibels A-weighted (dBA) minimum. Water seepage into timer body shall not exceed 0.2 mL (see 4.6.12.1).

3.6.3 Harnesses. Two harnesses shall be supplied with the apparatus, the waist strap assembly, and the harness assembly.

3.6.3.1 Waist strap assembly. The waist strap assembly shall be as shown on Drawing 803-5184270.

3.6.3.2 Harness assembly. The main harness assembly shall be as shown on Drawing 803-5184270.

3.7 Weight of apparatus. The weight of the apparatus shall not exceed 7.49 kilograms (16.5 pounds) without the canister (see 4.6.13).

3.8 Color. Where black is the specified color, it shall be in accordance with FED-STD-595 in a range, color 26081 to 26122. Other colors shall be as specified on Drawing 803-5184270.

3.9 Finish. Finishes shall be as specified on Drawing 803-5184270.

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3.10 Identification. The apparatus shall be identified by a nameplate and a serial number. The nameplate shall be located on the plunger assembly as shown on Drawing 803-5184270. The nameplate shall contain the following information: "Navy Oxygen Breathing Apparatus", Type A-4, the Julian date of manufacture, and National stock number. The serial number shall be attached to the wearer's side of the breastplate, to the wearer's right, as shown on Drawing 803-5184270. The combination valve shall be marked with a decal stating the following "Install This Side Up", in accordance with Drawing 803-5184270. A date code shall be imprinted on the facepiece, head harness, and breathing bag connector tube in accordance with Drawing 803-5184270.

3.11 Workmanship. The apparatus shall be uniform in quality and shall be free from irregularities, defects, or foreign matter that could affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2).

4.1.3 Certificate of compliance. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. When first article inspection is specified (see 6.2.1), four complete breathing apparatuses shall be provided. The sample apparatuses shall be subjected to the examinations and tests as specified in 4.3.1 using the same methods, materials, and equipment as those which shall be used during regular production. In order to avoid destruction of the apparatus, sample pieces of nitrile rubber, neoprene-coated fabric, and coated fiberglass cloth shall be used as specified in 4.4.2. If any sample fails any examination or test, it shall constitute failure of first article inspection. Any sample containing a critical defect (see 4.4.2.4) fails first article test. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.3.1 First article inspection procedure. First article inspection procedure shall be as specified in 4.3.1.1 and 4.3.1.2.

4.3.1.1 Examination and nondestructive tests. The four breathing apparatuses furnished for first article inspection shall be subjected to the examinations and tests specified in table IV.

TABLE IV. First article inspection (examinations and nondestructive tests).

Examination or test	Requirement	Test	Sample quantity
Apparatus weight	3.7	4.6.13	4
Visual and dimensional examination	3.2.5, 3.3, 3.4, 3.6, 3.8, 3.9, 3.10, 3.11	4.5.1, 4.5.2	4
Lens examination	3.6.1.2.1	4.5.3	4
Lens tests			4
Visual characteristics test	3.6.1.2.1	4.6.2.1	
Light transmission test	3.6.1.2.1	4.6.2.4	
Rubber irritation tests	3.2.2	4.6.1.2	5
Resistance to gas flow	3.5.1	4.6.3	4
Wear test	3.5.6	4.6.7	4
Leakage tests	3.5.2	4.6.9	4
Apparatus body test	3.6.2	4.6.9.1	
Facepiece test	3.6.1	4.6.9.2	
Relief valve test	3.6.2.2.1	4.6.9.3	
Breathing bag test	3.6.2.2	4.6.9.4	
Apparatus output	3.5.7	4.6.10	4
Breathing tube test	3.6.1.9	4.6.11	4
Timer loudness and duration test	3.6.2.3	4.6.12.2	4
Breathing bag volume test	3.6.2.2	4.6.14	4
Timer accuracy	3.6.2.3	4.6.12.3	4

4.3.1.2 For destructive tests. The four breathing apparatuses furnished for first article inspection shall be inspected in accordance with 4.3.1.1 and then be subjected to the tests specified in table V.

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TABLE V. First article inspection (destructive tests).

Test	Requirement	Test	Sample quantity
Lens tests			4
Cylinder power test	3.6.1.2.2	4.6.2.2	
Strength and impact resistance test	3.6.1.2	4.6.2.3	
Rubber compound test			5
oven aging	3.2.2, 3.2.3	4.6.1.1	
Corrosion	3.5.3	4.6.4	1
Impact test	3.5.5, 3.6.2.1.3	4.6.6	1
Water intrusion test	3.6.2.3	4.6.12.1	4
Shock test	3.5.4	4.6.5	1
Self-extinguishing test	3.2.4	4.6.8	5

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the visual and nondestructive examination specified in 4.4.2.1, 4.4.2.2, 4.4.2.3 and 4.4.2.4 and group A, B, C, and D tests. When specified in the contract or order, a quality conformance inspection report shall be prepared (see 6.2.2).

4.4.1 Lot. For the purposes of quality conformance inspection, a lot shall consist of all apparatuses produced under essentially the same conditions and offered for inspection at one time, but shall not consist of more than 1300 at any one time.

4.4.2 Sampling. Sampling shall be as specified in 4.4.2.1 through 4.4.2.8.

4.4.2.1 Sampling for rubber compound tests. For each lot of apparatuses, samples of facepieces, breathing tubes, and head harnesses shall be furnished for testing as specified in 4.6.1.1 and 4.6.1.2. In order to avoid unnecessary destruction of parts, the samples shall be pieces of rubber compound identical in chemical composition, equivalent in cure, and shall be prepared at the same time using identical methods as material used in the lot offered for delivery. For the physical characteristics tests specified in 4.6.1.1, five sample pieces shall be provided of each part's material and shall be 15.24 by 15.24 by 0.203 ± 0.03 cm (6 by 6 by 0.080 ± 0.005 inch). Dumbbell specimens in accordance with FED-STD-601 shall be cut from the sample pieces. For the irritation test in 4.6.1.2, five samples shall be provided of each part's material. These samples shall be 1.27 by 1.27 cm (0.5 by 0.5 inch) with one reasonably flat surface and shall have no sharp edges or corners.

4.4.2.2 Sampling for neoprene rubber test. For each lot of apparatuses, five samples of neoprene rubber shall be furnished for testing as specified in 4.6.1.1. The samples shall be neoprene identical in chemical composition, equivalent in cure, and shall be prepared at the same time using identical methods as material used in the lot offered for delivery. The sample pieces shall be 15.24 by 15.24 by 0.203 ± 0.013 cm (6 by 6 by 0.080 ± 0.005 inches). Dumbbell specimens in accordance with FED-STD-601 shall be cut from the sample pieces.

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4.4.2.3 Sampling for coated fiberglass cloth test. For each lot of apparatuses, five samples of coated fiberglass cloth shall be furnished for testing as specified in 4.6.8. In order to avoid unnecessary destruction of parts, the samples shall be pieces of coated fiberglass cloth identical in chemical and physical composition and shall be prepared at the same time using identical methods as material used in the lot offered for delivery.

4.4.2.4 Sampling for visual examination. A random sample of apparatuses shall be selected from each lot in accordance with MIL-STD-105, general inspection level I, for the examination specified in table VI. The acceptable quality level (AQL) for single tightened shall be 1.0 for major defects and 2.5 for minor defects. Any item in the sample containing one or more defects shall not be offered for delivery. If any sample contains a critical defect, the entire lot shall be rejected.

TABLE VI. Classification of defects.

Categories	Defects
Critical:	
1	Breathing apparatus incomplete, component parts missing or improperly assembled as specified.
2	Breathing tubes punctured.
3	Combination valve discs or seats defective or damaged.
4	Relief valve not operable as specified.
5	Breathing bag punctured.
6	Timer not operable as specified.
7	Lens surface not free of cracks, splits or holes.
8	Lens retaining ring does not retain lens to form a gas-tight seal.
9	Combination valves incorrectly assembled.
10	Facepiece buckles not operable or as specified.
Major:	
101	Rubber materials not free from pits, bumps, nonfills, porous areas, blisters, stickey areas, air traps, flow cracks, cut, holes, and tears.
102	Rubber facepiece thickness not as specified.
103	Breathing tube lengths and types not as specified.
104	Quick disconnect couplings on breathing tubes not as specified.
105	Head harness strap locations and type not as specified.
106	Speaking diaphragm assembly not as specified.
107	Optical properties of lens not as specified.
108	Breathing bag not as specified.
109	Canister guard and breastplate not as specified.
110	Retaining system not as specified.

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TABLE VI. Classification of defects. - Continued

Categories	Defects
Minor:	
201	Metal hardware composition not as specified.
202	Color of rubber not as specified.
203	Color of breathing bags not as specified.
204	Blooming of rubber not as specified.
205	Breathing apparatus not clean, free of dirt, dust, oil, grease, or extraneous foreign materials.

4.4.2.5 Sampling for group A tests. A random sample of apparatuses shall be selected from each lot in accordance with MIL-STD-105, inspection level S-3, AQL 4.0, for single tightened as specified in table VII. Any item in the sample containing one or more defects shall not be offered for delivery.

TABLE VII. Group A test.

Test	Requirement	Test
Apparatus weight	3.7	4.6.13
Visual and dimensional examination	3.2.5, 3.3, 3.4, 3.6, 3.8, 3.9, 3.10, 3.11	4.5.1, 4.5.2
Lens examination	3.6.1.2.1	4.5.3

4.4.2.6 Sampling for group B tests. A random sample of apparatuses shall be selected from each lot in accordance with MIL-STD-105, inspection level S-3, AQL 2.5, for single tightened as specified in table VIII. Any item in the sample containing one or more defects shall not be offered for delivery.

TABLE VIII. Group B tests.

Test	Requirement	Test
Lens tests		
Visual characteristics test	3.6.1.2.1	4.6.2.1
Light transmission test	3.6.1.2.1	4.6.2.4
Resistance to gas flow	3.5.1	4.6.3
Wear test	3.5.6	4.6.7
Leakage tests		
Apparatus body test	3.6.2	4.6.9.1
Facepiece test	3.6.1	4.6.9.2
Relief valve test	3.6.2.2.1	4.6.9.3
Breathing bag test	3.6.2.2	4.6.9.4
Apparatus output	3.5.7	4.6.10
Timer loudness and duration test	3.6.2.3	4.6.12.2
Breathing tube test	3.6.1.9	4.6.11
Breathing bag volume test	3.6.2.2	4.6.14
Timer accuracy test	3.6.2.3	4.6.12.3

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4.4.2.7 Sampling for group C tests. A random sample of apparatuses shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, AQL 1.5, for single tightened as specified in table IX. Any item in the sample containing one or more defects shall not be offered for delivery.

TABLE IX. Group C tests.

Test	Requirement	Test
Lens test		
Cylinder power test	3.6.1.2.2	4.6.2.3
Corrosion	3.5.3	4.6.4
Impact test	3.5.5, 3.6.2.1.3	4.6.6
Water intrusion test	3.6.2.3	4.6.12.1

4.4.2.8 Sampling for group D tests. A random sample of apparatuses shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, AQL 1.0, for single tightened as specified in table X. Any item in the sample containing one or more defects shall not be offered for delivery.

TABLE X. Group D tests.

Test	Requirement	Test
Lens test		
Strength and impact resistance test	3.6.1.2	4.6.2.3
Shock test	3.5.4	4.6.5

4.5 Examinations.

4.5.1 Visual. Sample apparatuses shall be visually examined for conformance to the requirements of 3.4, 3.6.1.1.1, 3.6.1.3, 3.6.1.6, 3.6.1.6.1, 3.6.1.8.1, 3.6.2.1.2, 3.8, 3.9, 3.10, and 3.11.

4.5.2 Dimensions. Sample apparatuses shall be examined for conformance to the requirements of 3.2.5, 3.3, 3.6.1, 3.6.2, 3.6.2.1.1, 3.6.2.1.3, and 3.6.3.

4.5.3 Lens examination. The lens shall be examined for surface imperfections and other markings as specified in 3.6.1.2.1. The lens shall be examined for scratches and striae by being placed in the axis of a magnifying lens system and illuminated from the back by an incandescent lamp placed to fully illuminate the lens. Striae will appear as dark lines; scratches will appear as illuminated lines.

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4.6 Tests. Tests shall be as specified in 4.6.1.1 through 4.6.14.

4.6.1 Rubber compounds tests. Rubber compounds tests shall be as specified in 4.6.1.1 through 4.6.1.2.3.

4.6.1.1 Physical characteristics. Rubber compound and neoprene rubber samples shall be selected as specified in 4.4.2.1 and 4.4.2.2, respectively, and shall be tested in accordance with ASTM D 412. Oven aging shall consist of exposing the samples to an air oven at 100 ± 1°C (212 ± 2°F) for 24 hours or at 70 ± 1°C (158 ± 2°F) for 168 hours. Samples of nitrile rubber shall conform to the requirements of 3.2.2 and samples of neoprene rubber shall conform to the requirements of 3.2.3.

4.6.1.2 Rubber irritation tests. Rubber compound samples shall be selected as specified in 4.4.2.1 and shall be tested for conformance to 3.2.2 by the following procedure.

4.6.1.2.1 Supervision. The rubber irritation tests shall be performed only under the supervision of a qualified physician in the presence of a designated Government representative.

4.6.1.2.2 Procedure. Five test subjects shall have a sample from each part (see 4.4.2.1) firmly affixed with surgical adhesive tape to the inner side of their upper arm. The samples shall be worn for 24 hours and then shall be removed. Upon removal of the samples, 24 hours after removal, and 48 hours after removal, the skin in the sample contact areas shall be examined for reactions as specified in table XI. If a plus 1 or higher reaction occurs during the 24-hour period when the samples are being worn, the test shall be immediately discontinued.

4.6.1.2.3 Test results. Any reaction of plus 1 or higher shall constitute failure of the rubber compound in the part or parts causing the reaction.

TABLE XI. Severity of skin reactions.

Rating	Reaction
0	No reaction
+ 1	Redness
+ 2	Redness and swelling
+ 3	Pimples and blisters
+ 4	Ulceration or tissue destruction

4.6.2 Lens tests. Lens tests shall be as specified in 4.6.2.1 through 4.6.2.4.

4.6.2.1 Visual characteristics test. Sample lenses shall be tested for resolving power by means of a telescope with a magnifying power of at least six, resolving at least 15 arc seconds, stopped down to an apperture of 10 mm (0.4 inch) in accordance with NBS Circular Special 374, test pattern 14, using a distance of 4.572 meters (15 feet). The lens shall conform to the requirements of 3.6.1.2.1.

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4.6.2.2 Cylinder power test. Sample lenses shall be tested for cylinder power through the use of a lensometer or other instrument. The lens axis shall be located and marked. The cylinder power shall be determined at sufficient locations on the surface to indicate the positions of the maximum values. The lens shall conform to the requirements of 3.6.1.2.2.

4.6.2.3 Strength and impact resistance test. Sample lenses shall be mounted normally in facepieces including the retaining rings. The facepiece shall be mounted on a wooden block so that the outer surface of the lens faces upward. The block shall be shaped to support the lens seal and the retaining rings, but the block shall not support the lens. The whole assembly shall be on a flat, horizontal surface. A 25.4-mm (1-inch) diameter steel ball weighing approximately 68 grams (2.4 ounces) shall be freely dropped from a height of 1.27 meters (50 inches) onto the center of the lens. The sample lenses shall conform to 3.6.1.2.

4.6.2.4 Light transmission test. The luminous transmittance of sample lenses shall be determined in accordance with ASTM D 1003. The transmission shall be as specified in 3.6.1.2.1.

4.6.3 Resistance to breathable gas flow. Sample apparatuses shall be tested for resistance to breathable gas flow which shall not exceed the requirement in 3.5.1. For the purpose of this test, a dummy canister with the known resistance of 18 mm (0.71 inch) water column height at a breathable gas flow rate of 85 liters per minute (L/min) (3 cubic feet per minute (ft^3/min)) shall be inserted into the apparatus. The supply of breathable gas shall be provided at the exhalation port in the facepiece and shall be at a flow rate of 85 L/min (3 ft^3/min).

4.6.4 Corrosion. Each sample apparatus shall be placed upright in a salt spray cabinet and shall be subjected to a salt spray atmosphere for 100 hours with a 5 percent salt solution in accordance with FED-STD-151. Sample apparatuses shall conform to 3.5.3.

4.6.5 High impact shock. Sample apparatuses shall be subjected to shock testing for grade A, class 1, type A lightweight equipment in accordance with MIL-S-901. The mounting fixture shall be an enclosed aluminum alloy box with internal dimensions of 72.1 by 21.6 by 59.5 cm (28-3/8 by 8-1/2 by 23-7/16 inches). The apparatus shall be placed in this container in accordance with Drawing 804-5773272 and shall remain in a horizontal position during testing. Sample apparatuses shall conform to the requirements of 3.5.4.

4.6.6 Impact. Sample apparatus bodies in both the standby and actuated positions shall be dropped onto a concrete surface. In the standby position an actual canister shall be inserted into the body, and in the actuated position a dummy canister that shall be of the same weight as an actual canister in accordance with MIL-C-17671 shall be inserted into the body. The lowest point of the body shall be 0.91 meter (3 feet) above the concrete in each orientation. The orientations from which each body in both positions shall be dropped shall be as follows:

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(a) Vertical with the retaining assembly at the lowest point.
(b) Inverted vertical with the retaining assembly at the highest point.
(c) Horizontal with the convex side of the canister guard at the lowest point.
(d) Horizontal with the convex side of the canister guard at the highest point.
(e) At 45 degrees from vertical on one side with the retaining assembly near the lowest point.
(f) Rotated 180 degrees from (e) with the retaining assembly near the highest point.

Sample apparatus bodies shall conform to the requirements of 3.5.5 and 3.6.2.1.3.

4.6.7 Wear. Sample apparatus bodies shall have a spent canister loaded and unloaded for 100 cycles. A cycle shall consist of inserting the spent canister in the standby position, moving bail to the actuated position, returning to the standby position, and removing the canister utilizing the tab in the retaining assembly. Sample bodies shall conform to the requirements of 3.5.6.

4.6.8 Self-extinguishing. The coated fiberglass cloth samples as specified in 4.4.2.3 shall be prepared and tested for flameout in accordance with FED-STD-191, test 5903. The cloth samples shall conform to the requirements of 3.2.4.

4.6.9 Leakage tests. Leakage tests shall be conducted in accordance with the procedures specified in 4.6.9.1, 4.6.9.2, and 4.6.9.3 to determine conformance to the requirements of 3.5.2.

4.6.9.1 Apparatus body procedure. Each sample apparatus body shall have a dummy canister inserted to form a gas-tight seal with the plunger assembly. The inhalation and exhalation tube fittings shall be sealed gas-tight. The relief valve shall be prevented from venting. The apparatus body shall be subjected to a pressure of 25.4 cm (10 inches) of water column height (manometer diameter shall be a minimum of 3.175 mm (0.125 inch)) to inflate the breathing bags. One minute after reducing the pressure to 12.7 cm (5 inches) of water column height, the pressure drop shall be checked for conformance to 3.6.2.

4.6.9.2 Facepiece assembly procedure. Each sample facepiece assembly shall have the inhalation and exhalation tube fittings sealed and shall be placed over a headform to form gas tight seal. A vacuum of 1.52 cm (0.60 inch) of water column height shall be applied to the inside of the facepiece. An in-leakage test material shall be applied to the outside of the entire facepiece assembly. The percent in-leakage shall be checked after 30 seconds into the test for conformance to 3.6.1.

4.6.9.3 Relief valve procedure. Each sample apparatus body shall have a dummy canister inserted to form a gas-tight seal with the plunger assembly and the inhalation and exhalation breathing tube fittings shall be sealed gas-tight. The relief valve shall be prevented from venting. The apparatus

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body shall be subjected to a gas flow which shall inflate the body to a pressure of 10.16 cm (4 inches) of water column height (manometer diameter shall be a minimum of 3.175 mm (0.125 inch)). The restraint shall be removed from the relief valve. The flow rate shall be checked while the pressure shall be maintained at 10.16 cm (4 inches) of water, for conformance to 3.6.2.2.1.

4.6.9.4 Bag volume. The breathing bags shall be drawn flat by applying a vacuum of 4 to 6 inches of water. The bags shall then be inflated and the volume measured. The volume in the bag shall be 5 liters (0.18 ft^3) minimum when the relief valve opens (see 3.6.2.2).

4.6.10 Performance. Following the tests in 4.6.3 and 4.6.9, each sample apparatus shall be subjected to a performance test. The performance test shall be as specified in 4.6.10.1 or 4.6.10.2. In both methods test samples shall be taken at the point where the breathable gas enters the facepiece after exiting the apparatus body. The sample apparatus shall conform to 3.5.7. The duration of operation shall be determined by the carbon dioxide levels exceeding the requirement of 3.5.7 or the breathing resistance exceeding 95 mm (3.74 inches) of water column height.

4.6.10.1 Automatic breathing metabolic simulator method. For this test, a canister shall be properly inserted into the sample apparatus. The facepiece shall be connected to the headform of an automatic breathing metabolic simulator. After actuating the canister, the apparatus shall be tested for at least 50 minutes using the following metabolic parameters:

(a)	Oxygen consumption rate	- 1.35 L/min (0.048 ft^3/min)
(b)	Carbon dioxide production rate	- 1.15 L/min (0.041 ft^3/min)
(c)	Ventilation rate	- 32.00 L/min (1.13 ft^3/min)
(d)	Respiratory frequency	- 22.0 breaths per minute
(e)	Exhaled humidity	- 80 percent saturated
(f)	Exhalation temperature	- 35°C (95°F)

4.6.10.2 Step up - step down method. For this test, a sample apparatus with a canister shall be placed on a test subject. The test shall be conducted as follows:

(a) Using a box 21.6 cm (8.5 inches) high, a subject shall step up on it and down from it at a rate that will produce an oxygen consumption rate of 1.35 L/min (0.048 ft^3/min) for 1 minute (work).
(b) Subject shall then rest for 1 minute.
(c) (a) and (b) shall be repeated for the duration of the test, except while taking test samples.
(d) A 2-minute period shall be allowed for taking test samples after each 10 minutes of working and resting.
(e) The duration of the test shall be not less than 50 minutes.

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4.6.11 Breathing tube. Each sample breathing tube with fitting shall be immersed in water and subjected to an internal air pressure of 34.5 kilopascals (kPa) (5 pounds per square inch gauge). No leakage shall occur while pressure is applied for a minimum of 10 seconds. The breathing tube shall conform to the requirements of 3.6.1.9.

4.6.12 Timer test procedures. Timer test procedures shall be as specified in 4.6.12.1 and 4.6.12.2.

4.6.12.1 Water intrusion. Each sample timer assembly shall be placed with knob up under a 1.59 cm (0.625 inch) inside diameter pipe and shall be subjected to water flow from that pipe for 60 minutes at a rate of 2.88 L/min (0.75 gallons per minute (gal/min)). At the conclusion of the hour, the sample timer L/min assemblies shall be inspected for conformance to 3.6.2.3.

4.6.12.2 Loudness. Each sample timer assembly shall be placed so a sound level meter is 30.5 cm (12 inches) directly above the bottom of the timer's knob. When the alarm bell is actuated, both the loudness and duration shall be measured. The bell shall perform to 3.6.2.3.

4.6.12.3 Accuracy. Each sample timer shall be wound, set at 1 hour, and activated. The timer shall perform to the requirements of 3.6.2.3.

4.6.13 Weight. Each complete sample apparatus shall be tested for conformance to 3.7.

4.6.14 Breathing bag volume. Each sample apparatus body shall have the plunger assembly and inhalation and exhalation tube fittings sealed gas-tight. The breathing bags shall be drawn flat by a vacuum of 12.7 cm (5 inches) water column height. The bags shall be inflated. The volume of gas required to just open the relief valve shall be measured and shall conform to 3.6.2.2.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein. Any sample not meeting the requirements of section 5 shall be cause for rejection of the entire lot.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

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5.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B - Type II - weather resistant.
Category I - general use.
Level C - Type I - non-weather resistant.
Category I - general use.

5.1.1.2 Fiberboard. Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index and specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The breathing apparatus (see 3.4) shall be cleaned and dried by a process and procedure that will insure that the apparatus is free of all petroleum products, hydrocarbons, and other foreign material that may affect the operation of the apparatus. Cleaning and drying shall be in accordance with MIL-P-116. Unless otherwise specified (see 6.2.1), the selection of the cleaning process and the drying procedure shall be at the contractor's option.

5.2.1.2 Unit protection. Each assembled (ready-for-use) breathing apparatus shall be individually unit protected within a transparent bag in accordance with MIL-P-116, method IC-1. Interior wrapping or cushioning material shall be provided, as necessary, to prevent striae and scratches to the facepiece lens, and puncture or damage to the protective greaseproof, waterproof bag. Each IC-1 unit bag shall be supplemented with a snug-fitting, class-weather-resistant fiberboard box conforming to PPP-B-636. Unless otherwise specified (see 6.2.1), box options such as type, variety, grade and style shall be the choice of the contractor. Cushioning or blocking shall be provided at the contractor's option. Use of loose-fill material is prohibited as cushioning, filler, stuffing or dunnage material. Cushioning or wrapping materials as applicable shall have properties (characteristics) for resistance to fire. Blocking shall be by the use of fiberboard forms and be of the same material as the unit box. Box closure and reinforcing shall be in accordance with the appendix to the box specification, method V. Unless otherwise specified (see 6.2.1), overpacking is not required for shipment.

5.2.2 Level C. Preservation shall be as specified for level A except that (a) the unit protection shall be method IC-3, bag closure by twist tie, or pressure-fit, and the leak and heat-seal tests are not required and (b) the unit fiberboard box shall be class-domestic/fire retardant (see 5.1.1.2) or when specified (see 6.2.1) class-domestic. Box closure shall be in accordance with the appendix to the box specification, method I, using pressure sensitive tape. Unless otherwise specified (see 6.2.1), overpacking in exterior shipping containers is not required for levels B, C, and commercial packing requirements.

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5.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning, and unit pack) shall be in accordance with ASTM D 3951.

5.3 Packing. Packing when required (see 6.2.1), shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities.

5.3.2 Levels A, B, and C containers. Breathing apparatuses preserved as specified (see 5.2) shall be packed when specified (see 5.3) in exterior shipping containers in accordance with table VII of MIL-STD-2073-1 for the level of packing specified (see 5.3) and herein. Unless otherwise specified (see 6.2.1), container selection including container options shall be the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2.1), level A shipping containers containing apparatuses preserved level C or commercial shall be provided waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Skidded containers. Wood, plywood, and cleated type containers exceeding 90.72 kg (200 pounds) gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1, appendix F or the applicable container specification and appendix thereto as applicable.

5.3.3 Commercial. Apparatuses preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 90.72 kg (200 pounds) gross weight shall be provided with a minimum of two 7.62 by 10.16 cm (3- by 4-inch) nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment and storage.

5.4 Palletized unit loads. When specified (see 6.2.1), containers shall be palletized in accordance with MIL-STD-2073-1.

5.5 Marking.

5.5.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1), and 3.10 interior (unit) packs, shipping containers and palletized unit loads shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

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6. NOTES

6.1 Intended use. The oxygen breathing apparatus is intended to be used with MIL-C-17671 quick-starting canister to provide fire fighting, damage control and other personnel with safe and sufficient breathing air for entry into spaces where the ambient air is contaminated with toxic gases or deficient in oxygen.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1 and 4.3).
(c) When fire-retardant requirements are not required (see 5.1).
(d) Level of preservation and packing required (see 5.2 and 5.3).
(e) Cleaning and drying selection if other than contractor's choice (see 5.2.1.1).
(f) Box options if other than contractor's choice (see 5.2.1.2 and 5.3.2).
(g) When overpacking is required (see 5.2.1.2, 5.2.2 and 5.3).
(h) When class-domestic box is required (see 5.2.2).
(i) When caseliners are not required (see 5.3.2.1.1).
(j) Palletization when required (see 5.4).
(k) Special marking required (see 5.5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Inspection system program plan	DI-R-4803	----
4.1.3	Certificate of compliance	DI-E-2121	----
4.3	First article inspection report	DI-T-4902	----
4.4	Inspection and test reports	DI-T-5329	----

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Conditions for use of level B preservation. This level of protection should be reserved for the acquisition of breathing apparatuses for resupply worldwide under known favorable handling, transportation and storage conditions.

6.6 Supersession data. This specification includes the requirements of MIL-B-17675B, dated 29 April 1966.

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6.7 Subject term (key word) listing.

Agerite (or VGB)
Firefighting equipment
Gas mask
Life support devices
Neoprene-coated fabric
Rescue equipment
Rubber compound
Self-extinguishing fabric
Skin tests, rubber irritation
Tubes

Custodians:
Army - EA
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

Preparing activity:
Navy - SH
(Project 4240-0544)

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